

PerfectMatchLFP26

Staff manual: suggesting international–local friend pairs

Local Friendship Programme 2026 ■ University of Jyväskylä

In one minute

PerfectMatchLFP26 reads the survey export and suggests one local friend for every international student whenever the rules allow it. Each pair is graded like a school report, from **1** insufficient to **5** perfect. You stay in control: change the weights or anyone's must-have, compute again, compare, and save the result. Everything runs on your own computer.

1. **Load** the survey CSV.
2. **Set** the weights (or keep the defaults).
3. **Check** the must-haves, especially the ones marked in red.
4. **Compute** and read the results.
5. **Improve**: lock good pairs, forbid unsuitable ones, adjust must-haves, compute again.
6. **Save** result.csv, report.txt, settings.csv and must_haves.csv.

1 Start and stop

1. Right-click **PerfectMatchLFP26-Windows.zip** and choose **Extract All**.
2. Open the extracted **PerfectMatchLFP26** folder and double-click **PerfectMatchLFP26.exe**. Keep the `_internal` folder next to it. Nothing needs to be installed.
3. Your web browser opens at **http://localhost:8766**. "Localhost" means your own computer; no internet connection is used.
4. When you have finished and saved your files, click **Close app** at the top right. Closing the browser tab alone does not stop the app.

First time? Click **Try fictional example** and follow this manual with it. The example has 20 internationals and 28 locals who do not exist. The screenshots in this manual show this example.

2 The screen at a glance

The **left side** is where you work (the *interact* phase). The **right side** shows the result of each computation (the *compute* phase). You switch between the two as often as you like.

The numbers and letters in Figure 1 are used throughout this manual.

3 Step by step

Step 1 ■ Load the survey

Export the survey results as **CSV** with **both heading rows** (the question row and the option row). Do not edit the export by hand. Click **Choose survey CSV** **1** and select the file. The app accepts both "CSV"

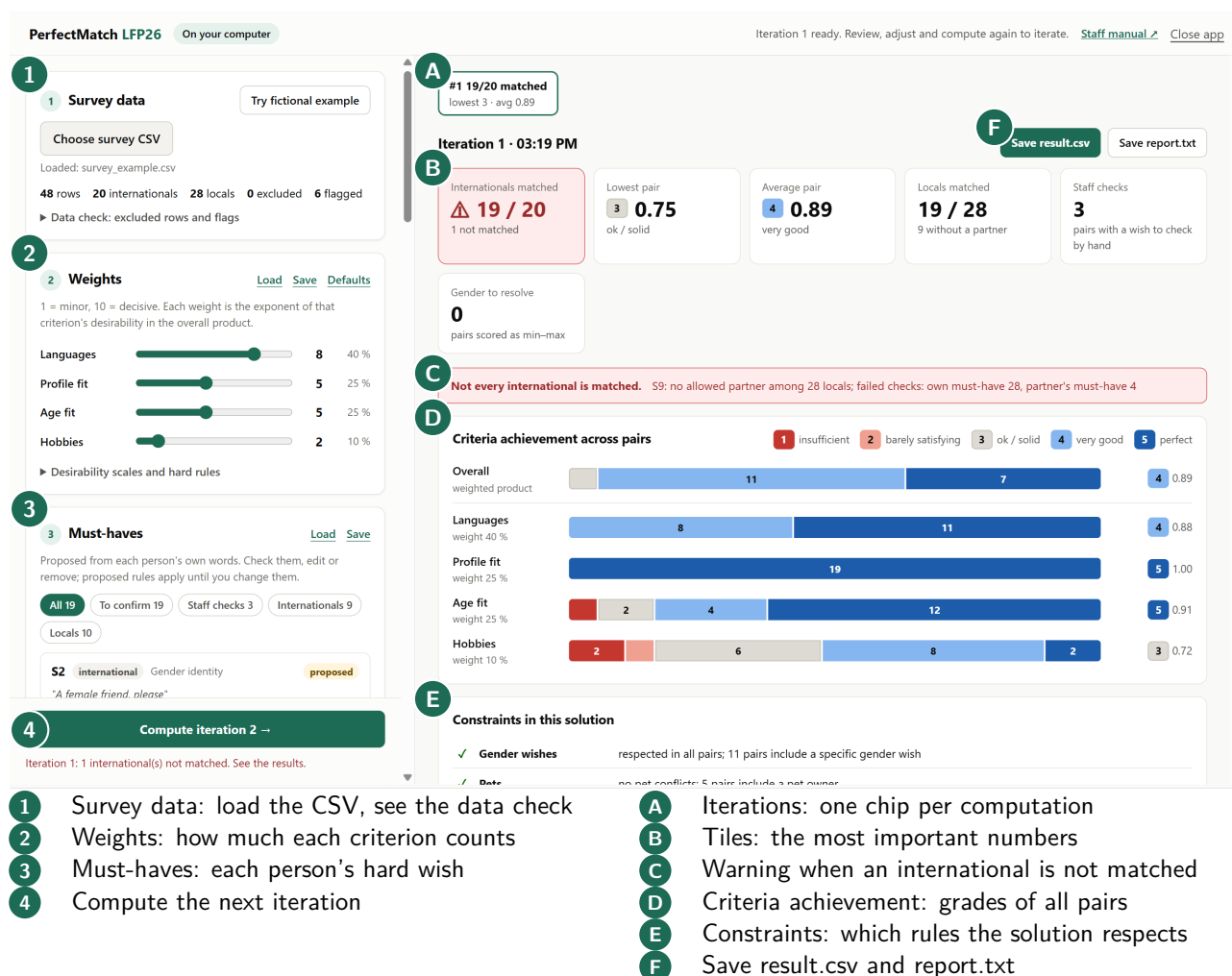


Figure 1: The workspace after the first computation of the fictional example.

and "CSV UTF-8" from Excel.

The line under the button tells you how many rows were read, how many internationals and locals are included, and how many rows were *excluded* or *flagged*. Open **Data check** to see why.

- **Excluded** (left out of the matching): the ID is missing or appears twice, the status is not clearly "local resident" or "international student", a consent statement was not accepted, the row is damaged, or more than 2 of the 8 key answers are missing.
- **Flagged** (included, but please look): for example a blank own gender ("to be resolved"), pet answers that contradict each other, or a must-have without an explanation.

Step 2 ■ Set the weights

2 Weights

1 = minor, 10 = decisive. Each weight is the exponent of that criterion's desirability in the overall product.

[Load](#) [Save](#) [Defaults](#)

Languages		8	40 %
Profile fit		5	25 %
Age fit		5	25 %
Hobbies		2	10 %

▼ Desirability scales and hard rules

Harrington desirability: the "just acceptable" value scores 50 %, the "fully satisfactory" value 80 %.

CRITERION	JUST ACCEPTABLE · 50 %	FULLY SATISFACTORY · 80 %
Shared language level	basic	good
Shared hobby categories	2	4
Age outside wish (%)	25	10
Missed profile wish counts (%)	25	

☒ Respect gender wishes (hard rule)
☒ No pet conflicts (hard rule)
☒ At least one shared language (hard rule)

Exclude rows missing more than ... of 8 key answers

Solver time limit per stage (s)

Four things decide how good a pair is. Move a slider from **1 (minor)** to **10 (decisive)**. The percentage on the right shows each criterion's share; the defaults are 40/25/25/10 %.

Languages the best language both speak, at the lower of the two levels

Profile fit does each person fit what the other asked for (one person, couple, family, ...)

Age fit is each person in the age range the other asked for

Hobbies how many of the 13 hobby categories they share

Desirability scales and hard rules holds the finer settings. The three **hard rules** are normally left on: respect gender wishes, no pet conflicts, at least one shared language.

Save writes **settings.csv**; **Load** reads it back; **Defaults** restores the starting values.

Step 3 ■ Check the must-haves

In the survey, everyone could name one *must-match* criterion and explain it in their own words. The app turns each answer into a rule and shows it as a card **3**.

S9 international Profile type proposed

"A family with a child aged 6-12"

Partner's profile is Family, children 5-15

no locals meet this

[Done](#) [Mark checked](#) [Remove](#)

Rule type Partner's profile is

Each card shows

- who it is and what they chose,
- their own words in quotes,
- the rule the app will apply,
- how many people on the other side meet it.

Red "no locals meet this" means the rule makes a match impossible. Read the person's words again: is the rule too strict, or does the wish simply not fit this round?

Use **Edit** to change the rule, **Mark checked** when it is right, **Remove** to drop it, and **Add a must-have** at the bottom for a wish you know about from elsewhere.

Tip Click the filter **To confirm** and go through the cards one by one. Editing a card marks it as checked.

Staff check. Some wishes cannot be checked from the survey data, for example an education level or “not from my own department”. These become a *staff check*: the app does not enforce them, but every pair that involves this person is marked 🚩 so that you can check it by hand.

Rule type	What it requires of the partner
Partner's gender is	a given gender
Partner speaks	one of the named languages, at least at the given level
A shared language at level	any language both speak, at least at that level
Partner has hobby	at least one (or a set number) of the named hobby categories
Shared hobby categories	at least this many hobby categories in common
Partner's profile is	one of the named profiles (e.g. Family, children 5–15)
Partner's age within	an age range such as 20–30
Partner's country of origin	one of the named countries
Partner has no	a given kind of pet (or no pets at all)
Staff check	nothing automatic; you check the pair by hand

Step 4 ■ Compute and read the results

Click **Compute iteration 1** ④. After a few seconds the right side fills.

Tiles ⑤. *Internationals matched* is the most important number: it is green with ✓ when everyone has a partner. If not, a red warning ⑥ names each international without a partner and explains why. *Lowest pair* and *Average pair* show the overall score with its grade. *Staff checks* and *Gender to resolve* count the pairs you should look at by hand.

Grades. Every score is between 0 and 1 and is turned into a grade:

Grade	Name	Score	Meaning
5	perfect	0.90 or more	as good as it gets
4	very good	0.80 or more	fully satisfactory
3	ok / solid	0.65 or more	a good basis
2	barely satisfying	0.50 or more	just acceptable
1	insufficient	below 0.50	look closely before confirming

Criteria achievement ⑦. One bar per criterion. Each coloured segment counts the pairs with that grade; point at a segment to see which pairs they are. The number on the right is the average.

Constraints in this solution ⑧. A tick confirms that a rule is respected in every pair. An exclamation mark asks you to look, for example at staff checks.

Step 5 ■ Look at each pair

Scroll down to **Pairs**. Each row shows the overall score and the grade of each criterion. Click a row to open both profiles side by side. Shared hobbies are highlighted, and each must-have shows whether it is met. A 🚩 line quotes a wish to check by hand (Figure 2).

- **Lock** keeps a pair you like in the next iteration.
- **Forbid** keeps a pair apart in the next iteration, for example after a staff check.

Step 6 ■ Improve: compute again

Every change on the left (a weight, a must-have, a lock or a forbid) is collected until you click **Compute iteration 2, 3, ...** Until then a yellow note says that the results do not include your changes yet.

Example. In the fictional example, iteration 1 matches 19 of 20 internationals. The red warning says that S9's own must-have rules out all 28 locals: S9 asked for “a family with a child aged 6–12”, and none of the locals is such a family. After talking to S9, you change the rule to a *staff check* and compute again.

Pairs Click a pair for both profiles. Lock keeps a pair in the next iteration; forbid excludes it.

PAIR	OVERALL	LANG.	PROFILE	AGE	HOBBIES	REVIEW
S6 – L18	4 0.88	4	5	4	5	Lock Forbid
S7 – L28	4 0.87	5	5	4	2	Lock Forbid
S8 – L19	4 0.88	4	5	5	4	1 Lock Forbid

Overall 4 0.88
very good

Languages 4 0.80
English, good

Profile fit 5 1.00
international's wish met;
local's wish met
international: grade 5 · local:
grade 5

Age fit 5 0.91
international's wish met;
local's wish missed by 7%
international: grade 5 · local:
grade 4

Hoobbies 4 0.88
5 shared: Music enjoying,
Reading and Writing,
Nature and Outdoor
Exploration, Travelling and
adventure, Socializing and
Entertainment

S8 · international

Country: Nepal

Profile: One person

Wishes profile: One person, Pair of friends, Small group of friends, Empty nester

Age: 36 - 40

Wishes age: 36 - 40, 41 - 45, 46 - 50, 51 - 55

Gender: Male

Wishes gender: all are fine

Languages: English (good), Finnish (basic), Nepali (native/very good)

Hobbies: Sports and Outdoor Activities, **Music enjoying**, **Reading and Writing**, **Nature and Outdoor Exploration**, **Travelling and adventure**, **Socializing and Entertainment**, Personal Development and Learning, Animal and Nature-Related Interests, Handicrafts, Cultural heritage

Pets: all pets welcome

Motivation: Learning about Finland

Appreciates: Open mind

L19 · local

Country: Suomi

Profile: One person, Native Finn

Wishes profile: One person, Pair of friends, Family, children under 5, Family, children 5-15

Age: 46 - 50

Wishes age: 41 - 45, 46 - 50

Gender: Male

Wishes gender: all are fine

Languages: English (good), Finnish (native/very good), German (good)

Hobbies: **Music enjoying**, Music playing, **Reading and Writing**, **Nature and Outdoor Exploration**, **Travelling and adventure**, **Socializing and Entertainment**, Visual Art

Pets: no pets at home; all pets welcome

Must-have: Other: "Not a student from my own department"

Motivation: Practising languages

Appreciates: Reliability

1 L19: check by hand - "Not a student from my own department"

PAIR	OVERALL	LANG.	PROFILE	AGE	HOBBIES	REVIEW
S9 – L24	4 0.85	4	5	5	2	1 Lock Forbid

Figure 2: The pairs table with one pair opened: grades per criterion, both profiles, and a wish to check by hand.

Iteration 2 matches all 20 (Figure 3).

Each computation adds a chip at the top **A**, with the number matched, the lowest grade, the average and how many pairs changed. Click an older chip to look at it again; **Restore its settings** brings back exactly the weights, must-haves, locks and forbids of that iteration.

Step 7 ■ Save

File	Where	Content
result.csv	Save result.csv F	one row per pair: scores, grades, locks, notes
report.txt	Save report.txt F	summary, grades, pairs, who is not matched and why, review notes, settings
settings.csv	Weights: Save	the weights and settings, to repeat the run
must_haves.csv	Must-haves: Save	the must-haves as you checked them

Your browser puts the files in its Downloads folder. Keep them together with the survey export in a dated folder. To continue later, load the survey, then **Load** the settings and the must-haves.

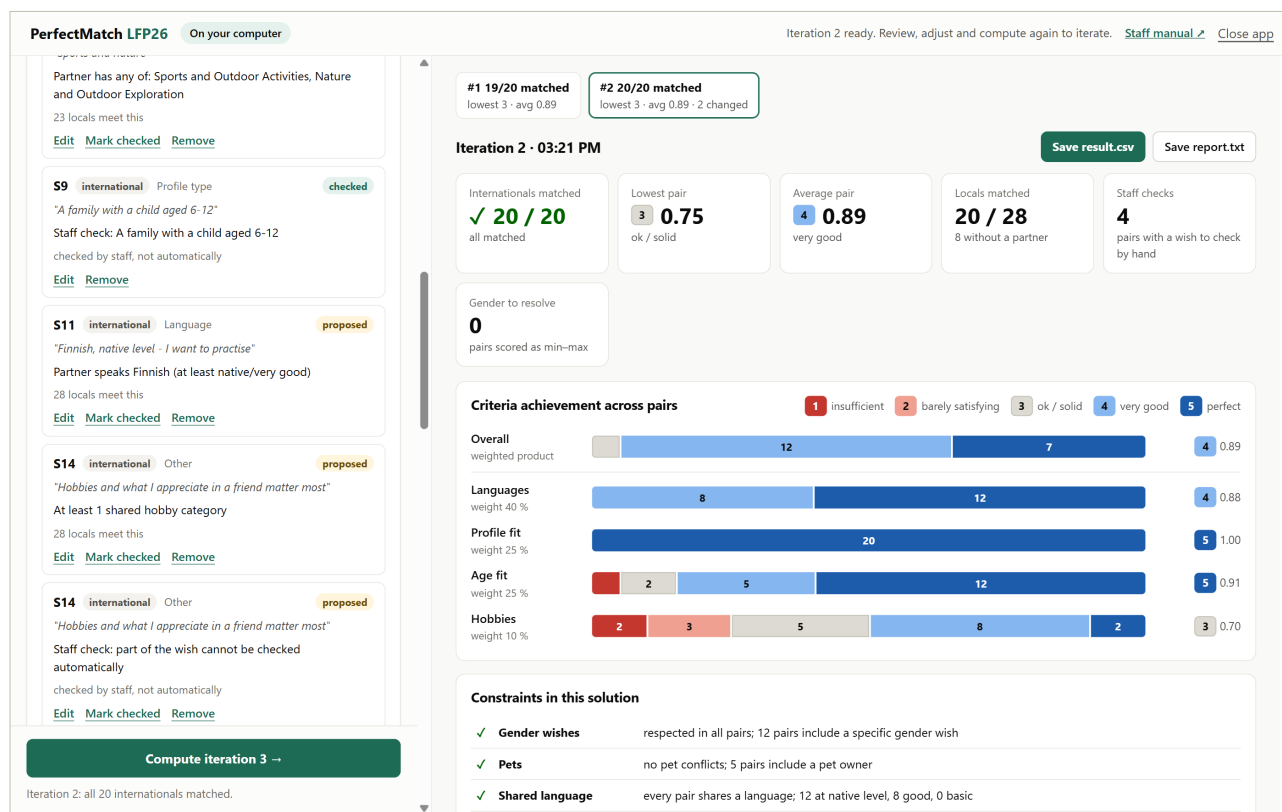


Figure 3: Iteration 2: S9's rule is now a staff check, and all 20 internationals are matched.

4 How the scores work

- Each criterion gets a score from 0 to 1. **0.50 means just acceptable, 0.80 fully satisfactory.** The defaults: a shared language at *basic* level is just acceptable and *good* is fully satisfactory; 2 shared hobby categories are just acceptable and 4 fully satisfactory; an age 25 % outside the wished range is just acceptable and 10 % outside fully satisfactory (inside the range is perfect); a missed profile wish counts 25 %.
- Profile and age are scored from both sides; the pair details show the international's and the local's view.
- The overall score **multiplies** the criteria, with the weights as powers. One weak criterion therefore pulls the whole pair down instead of being hidden by an average.
- The app first matches **as many internationals as possible**, then makes the **lowest pair as good as possible**, then the **total**. Locals may be left without a partner when there are more locals.
- Blank own gender.** If someone left their own gender empty and the partner wished for a specific gender, the pair is shown with a minimum (0, if it does not fit) and a maximum (if it fits). The app uses the average and counts the pair under *Gender to resolve*.

5 When something goes wrong

Message or problem	What to do
"This does not look like the survey export"	Export the survey again as CSV, with both heading rows.
"The survey export is missing these questions"	Export all questions, not a selection.
An international is not matched (red warning)	Read the reason. Relax that person's must-have (or make it a staff check), remove a lock or forbid, or check the gender and pet answers. Then compute again.
"no locals meet this" on a card	The rule is too strict or misread. Edit it.
"Cannot reach the app"	The app was closed. Start PerfectMatchLFP26.exe again and reload the page.
"This session has changed"	The app was restarted. Reload the page and load the survey again.
"The solver could not finish"	Increase the solver time limit in Desirability scales and hard rules .
The browser did not open	Open the address shown in your browser, usually http://localhost:8766 .
Windows blocks the app	The app is not signed. Ask IT to review it.

6 Data protection

Personal data The survey export, report.txt, result.csv and must_haves.csv contain participants' IDs and their own words. Store them like the survey export, in a protected folder, according to the programme's data protection rules, and delete them when they are no longer needed. Do not send them by e-mail without protection.

The app runs only on your computer. It does not use the internet, does not store the survey anywhere, and keeps the data in memory only until you close it. Programme staff review every suggestion before confirming a pair.